Democratization theory

A game-theoretic approach

Lecture 07

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- Last lecture: Two group model
- **Hypothesis**
  - Two groups: poor and rich
  - Normalize population to 1
  - δ: fraction of rich people
  - $1-\delta$: fraction of poor people
  - θ: share of the income of the rich people
  - y^p and y^r are rich and poor incomes
  - $y^p = \frac{(1-\theta)\bar{y}}{1-\delta}$ and $y^r = \frac{\theta\bar{y}}{\delta}$
  - Restrictions: $\delta < 1/2$, $y^p < \bar{y} < y^r$ (which implies that $\theta > \delta$)
  -
  - Problem of individual i (can be r for rich or p for poor)

$$\tau^i = \arg\max_{\tau \in [0,1]} V(y^i|\tau)$$

Adding more structure to the model

- Revolution constraint: let μ be a parameter that captures destruction of part of the resources if revolution is made

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- Payoff in case o revolution[1]: $V^p(R, \mu) = \frac{(1-\mu)\bar{y}}{1-\delta}$
- Revolution is attractive if $V^p(R, \mu) > V^p(y^p|\tau^N = \tau^r)$ which implies that $\theta > \mu$

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In short:

- Rich people: $V^r(y^p|\tau^N) = \frac{\theta\bar{y}}{\delta}$ and $V^r(R,\mu) = 0$
- Poor people: $V^p(y^p|\tau^N) = \frac{(1-\theta)\bar{y}}{1-\delta}$ and $V^p(R,\mu) = \frac{(1-\mu)\bar{y}}{1-\delta}$
- Therefore, theres is a conflict: rich prefer nondemocratic regimes and poor have an incentive to make a revolution

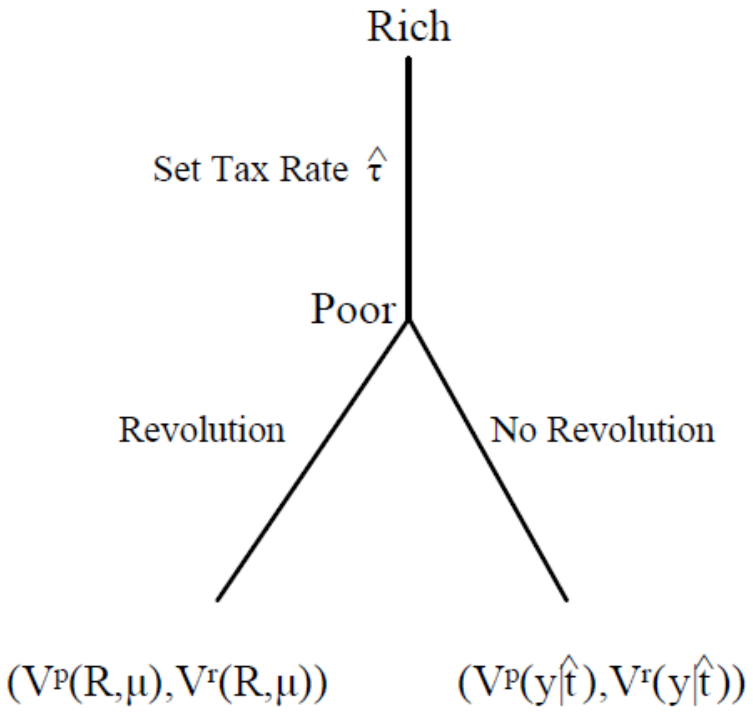

Figure: Democratization game - first part

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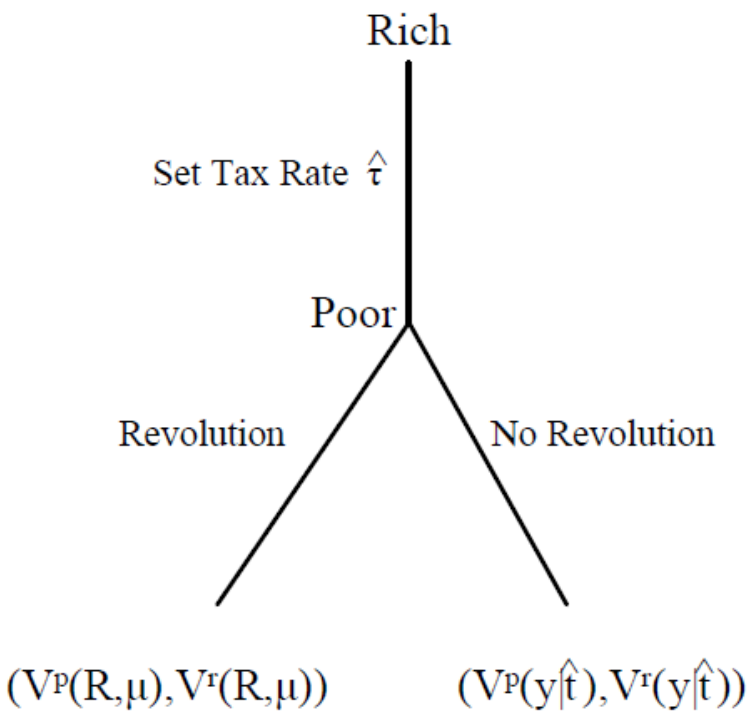
- **Formally:** a decision node initiates a subgame if neither it nor any of its successors are in an information set that contains nodes that are not successors to it.
- **Definition (SPNE):** A Nash equilibrium is said to be subgame perfect if an only if it is a Nash equilibrium in every subgame of the game.



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In order to avoid revolution, rich people can set a tax $\hat{\tau} > 0$

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  - Which tax will be chosen?
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- Suppose $\theta > \mu$:
  - In order to avoid revolution, $V^p(y^p|\tau^p) \geq V^p(R, \mu)$ must hold
  - That means: $\mu \geq \theta - (\tau^p(\theta - \delta) - (1 - \delta)C(\tau^p))$
  - If this last inequality holds, then rich people can choose a tax to avoid revolution
  - Tax chosen: $\mu = \theta - (\hat{\tau}(\theta - \delta) - (1 - \delta)C(\hat{\tau}))$
  - Note that $\hat{\tau} \leq \tau^p$

Proposition: There is a unique subgame perfect equilibrium in the game described before, and it is such that:

- If $\theta \leq \mu$, $\tau^N = 0$ and poor **do not** revolt.
- If $\theta > \mu$ and $\mu < \theta - (\tau^p(\theta - \delta) - (1 - \delta)C(\tau^p))$, then poor revolt.
- If $\theta > \mu$ and $\mu \geq \theta - (\tau^p(\theta - \delta) - (1 - \delta)C(\tau^p))$, then $\tau^N = \hat{\tau}$ where $\hat{\tau}$ is given by $\mu = \theta - (\hat{\tau}(\theta - \delta) - (1 - \delta)C(\hat{\tau}))$, and poor **do not** revolt.

Commitment Problems

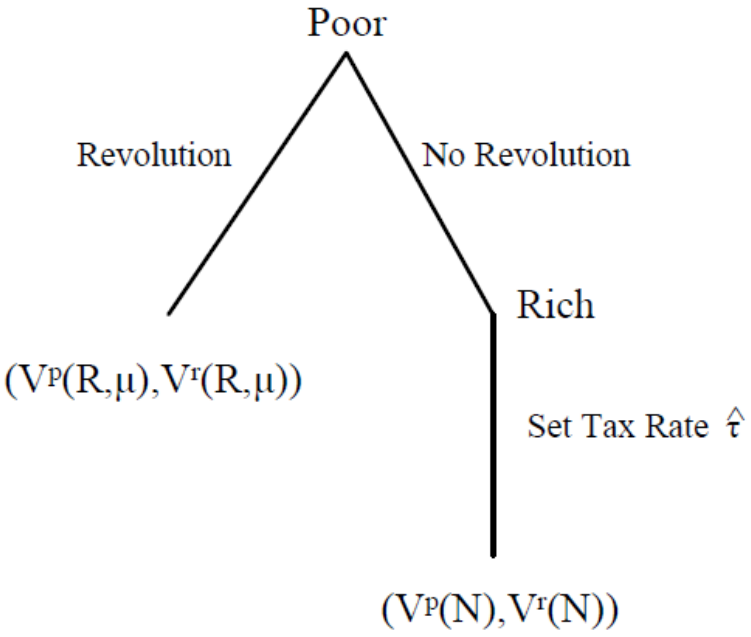

Figure: Commitment problem

Let's use backward induction to enunciate the equilibrium

Proposition: There is a unique subgame perfect equilibrium in the game, and it is such that:

- If $\theta \leq \mu$, then $\tau^N = 0$ and poor **do not** revolt.
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- Let's add these in the game

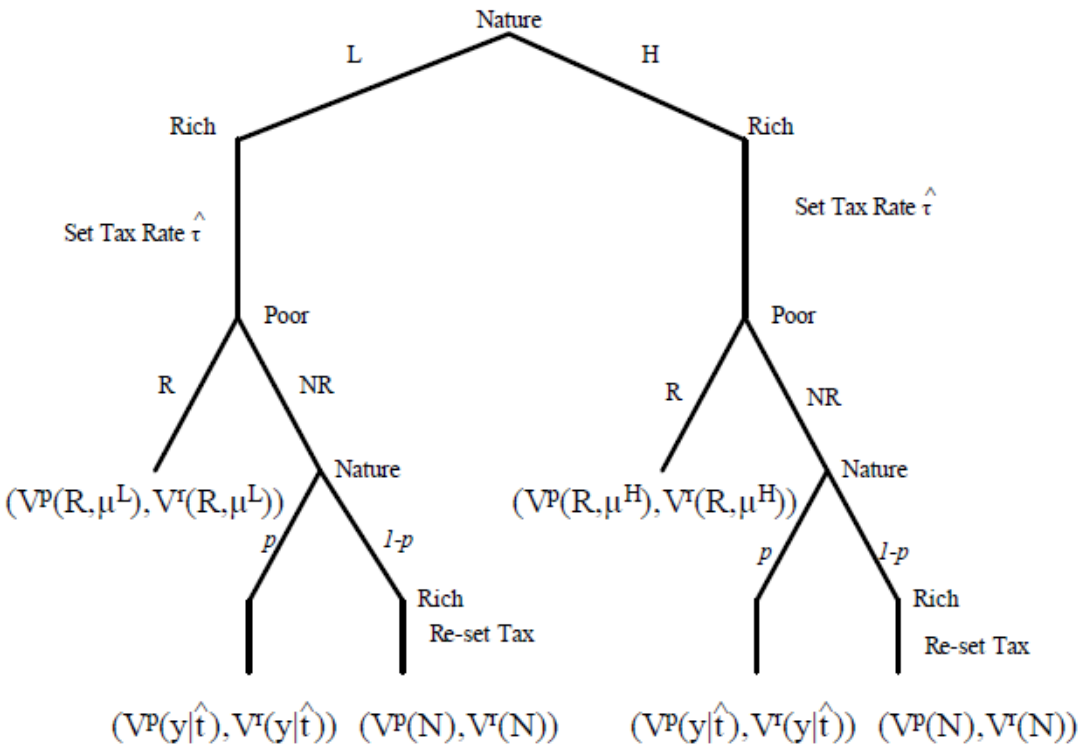

Figure: Game of Promises

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- Rich people payoff (revolution): $V^r(R, \mu^S) = 0$

- Focus: $S = H$
- If $V^p(R, \mu^H) > V^p(N)$, revolution threat is real ($\theta > \mu^H$)
- What tax the elite will offer to dissipate the threat?
- $V^p(N, \tau = \hat{\tau}) \geq V^p(R, \mu^H)$. This means:

$$y^p + p(\hat{\tau}(\bar{y} - y^p) - C(\hat{\tau})\bar{y}) \geq \frac{(1 - \mu^H)\bar{y}}{1 - \delta}$$

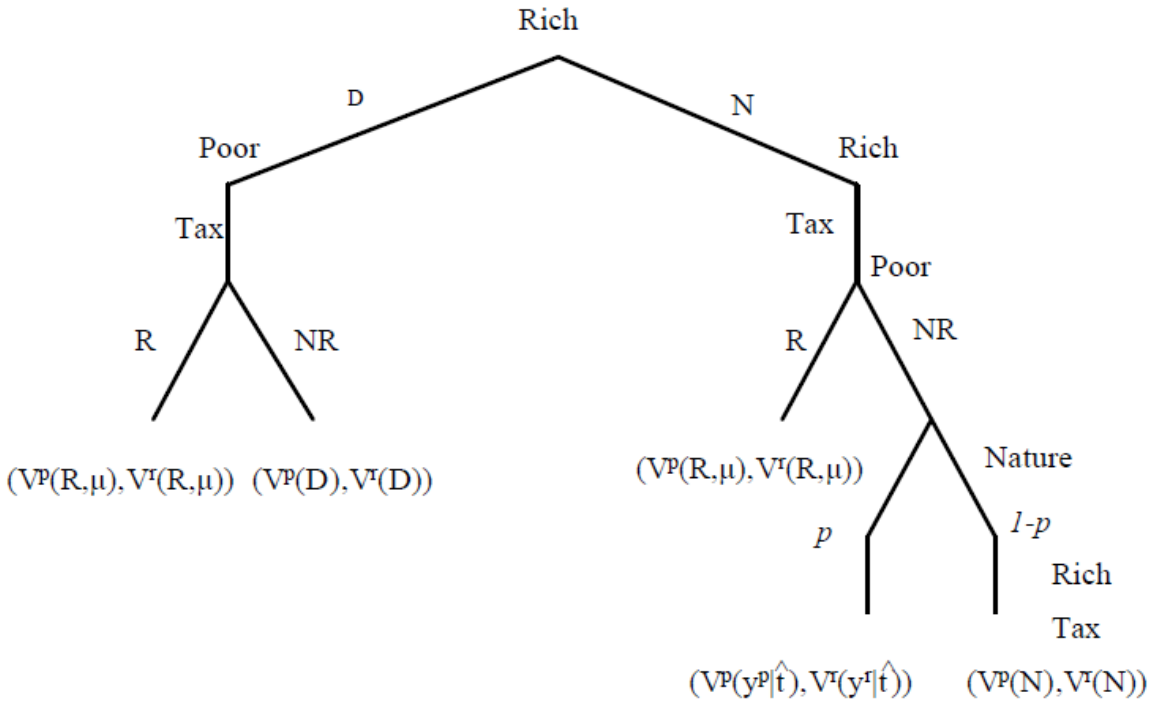
or

$$\mu^H \geq \theta - p(\hat{\tau}(\theta - \delta) - (1 - \delta)C(\hat{\tau}))$$

- Maximum commitment is $\hat{\tau} = \tau^p$
- Defining $\mu^* = \theta - p(\tau^p(\theta - \delta) - (1 - \delta)C(\tau^p))$

Proposition: There is a unique subgame perfect equilibrium in the game described before, and it is such that:

- If $\theta \leq \mu^H$, $\tau^N = 0$ and poor **do not** revolt.
- If $\theta > \mu^H$, constraint is binding in the high state. In this case:
  - If $\mu < \mu^*$, promises by the elites are not credible to avoid revolution.
  - If $\mu \geq \mu^*$, promises are credible and can avoid revolution.



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Next lecture:

- More on theory of democratization
- Revolution or repression as options
- Coups and consolidation